Rafal P. Wiewiora

rafal.p.wiewiora@gmail.com | LinkedIn | Google Scholar | Website

Summary

Molecular designer with roots in experimental chemistry and chemical biology, deep experience in large-scale molecular simulation, and fluency across the physics-based and ML models used in drug design. I lead in silico design for a small-molecule allosteric program across the full discovery arc, from target assessment to lead optimization, benchmarking every prediction against experiment. Much of my work establishes where these methods genuinely advance design and where they mislead: fine-tuning ADME models on internal data, extracting chemical interpretability, and evaluating structure-prediction tools (OpenFold, Boltz) while automating both the design of candidates and the structural inputs they are scored against. Increasingly I automate the entire design loop with agentic AI—building a reusable evaluation platform spanning SAR analysis, model-accuracy tracking, and generative chemistry that lets one scientist orchestrate an agent team with the reach of a traditional CADD group.

Education

PhD, Chemistry, Weill Cornell Medicine (Chodera Lab, Memorial Sloan Kettering) 2014 – 2020
MChem, Chemistry, University of Oxford 2010 – 2014

Experience

Principal Scientist & In Silico Design Lead 01/2023 – present
PsiThera (formerly Psivant Therapeutics)

- Lead computational designer on a small-molecule allosteric drug program, driving hit finding through lead optimization. Run weekly design cycles—enumerate and generate compounds with agentic AI, score with free energy and ML property models, select for synthesis, benchmark against experiment, and translate the hit rate into make/skip rules the team acts on that week. Across the broader TNF superfamily, ran allosteric-tractability assessment (MD and co-folding), virtual screens, and end-to-end fragment screening for two further targets—now extending into peptide design
- Built an AI-augmented design platform: automated SAR and structural-biology apps, model-accuracy dashboards with agentic explainability for outliers, and agent teams driving generative chemistry under expert medicinal-chemistry feedback. Overnight docking and simulation assistants turn design ideas into 3D poses; chemists weigh in by morning, closing the make–test–analyze loop seamlessly. Developing agentic molecular generation with RL reward shaping, surfacing capability gaps in chemical reasoning and SAR interpretation
- Evaluate structure-prediction tools (OpenFold, Boltz) for design-relevant accuracy. Exploring inference-time steering of open-source diffusion models with physics-informed guiding potentials—anchored by hands-on work forcing them from the symmetric TNFα state into the asymmetric one that drives allostery (with MD of those transitions)—toward conformational discovery without templates
- Benchmark co-folding on design-relevant perturbations (scaffold hops, bioisosteres), fine-tuning on both affinity and structure as a faster successor to FEP. Generate synthetic training data—via STORMM (PsiThera's GPU simulation engine), bioisostere enumeration, and RL-driven generation with reward models trained on medicinal chemists' 3D preferences—toward autonomous molecular design
- Assess and improve every predictive method: a weekly RBFE reliability assessment across multiple methods, distilling sources of error into actionable chemical guidance; ADME models directed end-to-end on internal data, with an uncertainty-based active-learning policy and drift monitoring
- Analyze large simulation datasets with NMR to develop mechanistic hypotheses for differential inhibition—using ML dimensionality reduction to extract the protein motions distinguishing active from inactive binders, interpreting cellular SAR into design guidance. Propose targeted mutations to validate findings experimentally

Senior Investigator, Computational Biophysics 06/2021 – 12/2022
Roivant Discovery

- Developed simulation methods to predict the 3D structures of drug-induced ternary complexes before crystal structures were available, extracting design rules for heterobifunctionals (PROTACs) and molecular glues (*JACS Au*, *Nature Comm.*); integrated with experimental hydrogen–deuterium exchange and NMR. Drove a KRAS hit-to-lead program spanning both inhibitor and PROTAC modalities
- Led a team to find hidden, covalently targetable pockets at proteome scale, combining the AlphaFold structural database with physics-based models

Computational Chemist, *Redesign Science* 02/2020 – 05/2021

- Helped build the first exascale molecular simulation campaign (280K GPUs), discovering hidden binding sites across the SARS-CoV-2 proteome (*Nature Chemistry*)

PhD Research, Chodera Lab, Memorial Sloan Kettering 07/2015 – 07/2020

- Generated models of large datasets of molecular simulation data to map how cancer-relevant enzymes change shape; then published a critical assessment of the field's standard analysis method, showing where it works and where it misleads (*eLife*, *JCTC*)
- Contributor to OpenMM and Folding@home; secured $228K DoD grant as PI

Technical Skills

AI & Agentic Tools Agentic Workflows · Evaluation Design · RL for Molecular Generation · Active Learning

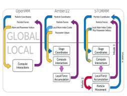
Drug Design & Simulation FEP · ADME Prediction · Automated SAR Analysis · Virtual Screening · Enhanced Sampling · Markov State Models · OpenMM · STORMM

Machine Learning Structure Prediction · Co-folding · OpenFold · Guided Molecular Generation · Generative Fine-tuning · QM/NNPs in Design · Model Benchmarking

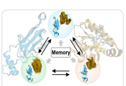
Biophysics & Experiment NMR · Cryo-EM · Hydrogen–Deuterium Exchange · Synthetic Chemistry · Chemical Biology

Software & HPC Python · Docker · RDKit · Dash / Streamlit · Folding@home · Cloud / GPU Clusters · Interactive Design Dashboards

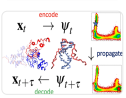
Selected Publications (13 total; * = co-first author)



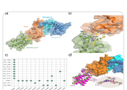
Cerutti, **Wiewiora** et al. (2024). STORMM: GPU simulation engine for chemical simulations. *J. Chem. Phys.*
Open-source C++/CUDA engine running thousands of parallel calculations per GPU



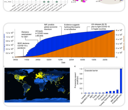
Qiu, **Wiewiora** et al. (2024). Non-Markovian models for drug-induced protein complex design. *JACS Au*.
Predicted protein-protein interfaces later experimentally validated



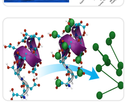
Jones, McDargh, **Wiewiora** et al. (2023). Molecular latent space simulators for distributed trajectories. *J. Phys. Chem. A*, 127, 5470–5490.
Learned latent-space dynamics generating continuous synthetic trajectories from distributed data; applied to PROTAC design



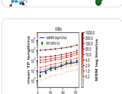
Dixon … **Wiewiora** … Izaguirre (2022). Predicting structural basis of targeted protein degradation. *Nature Comm.*, 13:5884.
Combining simulation with experiments to predict drug-induced complex structures



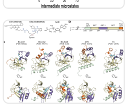
Zimmerman … **Wiewiora** … Bowman (2020). SARS-CoV-2 simulations go exascale. *Nature Chemistry*, 13, 651–659.
First exascale molecular simulation; discovered hidden binding sites across viral proteome



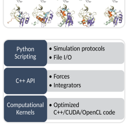
Gkeka … **Wiewiora**, Lelièvre (2020). ML force fields and coarse-grained variables in molecular dynamics. *JCTC*, 16, 4757–4775.
Review: how neural networks can learn to replace physics-based molecular energy models



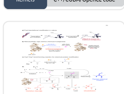
Wiewiora*, Suárez* et al. (2020). What Markov state models can and cannot do. *JCTC*, 17, 3119–3133.
Critical assessment of the field's standard analysis method—now a reference for best practices



Wiewiora*, Chen* et al. (2019). Dynamic conformational landscapes of SETD8. *eLife*, 8, e45403.
Milliseconds of GPU simulation revealing 24 hidden states of a cancer-relevant enzyme



Eastman … **Wiewiora, R.P.** … Pande (2017). OpenMM 7. *PLOS Comp. Biol.*, 13, e1005659.
A widely-adopted GPU-accelerated molecular simulation engine



Wright … **Wiewiora** … Davis (2016). Post-translational mutagenesis. *Science*, 354, aag1465.
Experimental chemistry: a new method for engineering protein building blocks